

Prateek Arora

Postdoctoral Researcher, Department of Civil and Environmental Engineering, University of California, Berkeley

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SUMMARY

Prateek Arora, Ph.D. in Civil Engineering, specializes in structural engineering, probabilistic hazard modeling, and disaster risk analysis. Prateek is committed to advancing methods for assessing natural hazard risks, informing resilient strategies for critical infrastructure, and supporting effective emergency responses. Prateek brings experience in research, teaching, consulting, and cross-disciplinary collaboration to develop resilient infrastructure against extremes.

ACADEMIC APPOINTMENTS

University of California, Berkeley, CA

August 2025 - Present

Postdoctoral Researcher, Disaster Risk Analysis Lab

Advisor: Dr. Luis Ceferino

EDUCATION

New York University, NY, Ph.D. in Civil Engineering

August 2025

Thesis: Probabilistic Modeling for Power Outages and Adoption of Renewables for Enhancing Power Infrastructure Resilience to Extreme Events

Advisor: Dr. Luis Ceferino

NYU Urban Doctoral Fellow, NYU School of Engineering Fellow

Stanford University, CA, M.S. in Structural Engineering

March 2020

KC Mahindra scholarship for graduate studies

BITS Pilani, Pilani Campus, India, B.E. (Hons.) in Civil Engineering

June 2018

Ranked 1st in the Civil Engineering class of 90 students

RESEARCH EXPERIENCE

University of California, Berkeley

August 2025 - Present

Postdoctoral Researcher, Disaster Risk Analysis Lab, Advisor: Dr. Luis Ceferino

Berkeley, CA

- Building a power outage risk model for Pacific Gas and Electric Company to assess earthquake impacts and design resilience strategies using grid infrastructure data, environmental variables, and historical failure data
- Developing digital-twin-based optimization frameworks to evaluate post-disaster repair strategies, resource allocation, and retrofit prioritization under uncertainty using multi-fidelity surrogates simulation, statistical modeling, Bayesian inference, and scenario-driven performance metrics
- Built workflows to collect, clean, and prepare multi-source datasets for modeling, including geospatial, infrastructure, environmental, and hazard data
- Studied recovery trajectories across events to estimate how outage intensity influences restoration rates, applying time-series analysis and comparative statistical modeling to quantify restoration dynamics
- Collaborated with the engineering team of a utility company to translate resilience questions into modeling tasks, identify measurable metrics, and define model evaluation criteria.

New York University

Jan 2021 - August 2025

Graduate Research and Teaching Assistant

New York, NY

- Developed probabilistic methods for quantification of hurricane-induced power outages during early warning by analyzing power outage data with more than 3 million records, determining hazard fields through hurricane models, and incorporating infrastructure attributes
- Developed a performance-based probabilistic framework to model risk to power systems during future hurricanes
- Conducted research on the resilience and economic benefits of the adoption of renewables during a hurricane emergency
- Implemented OpenMPI-based parallel workflows on the high-performance computing platform to run large-scale Monte Carlo simulations, quantifying recovery trajectories of energy communities under diverse hurricane scenarios
- Conducted wind hazard sensitivity and causal analyses to quantify how exposure and structural characteristics affect failure probability
- Performed geo-spatial wind impact analyses using satellite night-lights and building data to detect and validate outage-affected areas following hurricanes

- Selected as Coalition of Disaster Resilient Infrastructure Fellow; 15 out of 160 selected research proposals
- Published research in peer-reviewed journals as first author; presented at an international conference and an invited talk; collaborated with utilities and researchers
- Taught a graduate class of 13 students and an undergraduate class of 16 students
- Mentored 4 undergraduate and 7 graduate students on various research projects related to disaster risk analysis

CSIR-Structural Engineering Research Centre

Research Intern, Structural Health Monitoring Laboratory

July 2017 - December 2017

Chennai, India

- Applied robust statistical methods to account for environmental variability in damage localization.
- Published research article in a peer-reviewed journal on damage identification in bridge structures with limited sensors.

PROFESSIONAL EXPERIENCE

Rivera Consulting Group, Inc.

Staff Engineer

May 2020 - December 2020

San Francisco, CA

- Designed structural connections and coordinated with architects and building engineers for design requirements.
- Conducted analysis for retrofit of existing Steel-Frame, Unreinforced Masonry, and Wood-Frame Buildings.

Thornton Tomasetti, Inc.

Structural Engineering Intern

June 2019 - September 2019

San Francisco, CA

- Prepared reports for gravity and lateral loading for high-rise buildings according to building codes.
- Reviewed structural shop drawing submittals from contractors to ensure the structural requirements.

Mormugao Port Trust

Summer Intern, Civil Engineering Department

May 2016 - July 2016

Goa, India

- Performed costing and estimation for a railway signaling building at Mormugao Port.
- Prepared the report of construction, which included placement of formwork and detailing of steel reinforcement.

HONORS, AWARDS, AND ACHIEVEMENTS

Best Paper Award, ASCE EMI Objective Resilience Student Competition

Recognized for research leveraging AI and remote sensing for power infrastructure resilience.

2025

Coalition of Disaster Resilient Infrastructure Fellowship, \$15,000

Resilience of Power Grid with Renewables to Hurricanes in a Changing Climate.

2022

New York University Tandon School of Engineering Fellowship

Doctoral Studies in the field of Disaster Risk and Resilience of Urban Environments.

2021

New York University Urban Doctoral Fellowship, \$3,000

Fellowship awarded to 12 Doctoral Students across NYU working towards the development of Urban Cities.

2021

K.C. Mahindra Fellowship, \$2,000

Fellowship to 82 out of 1200 applicants from India to pursue graduate studies in the USA.

2018

O.P. Jindal Engineering and Management Scholarship

Awarded for academic and leadership excellence.

2016

Merit Scholarship, BITS Pilani

Awarded to the top 3% of students across all disciplines.

2017-18

National Standard Examinations in Chemistry (INChO)

Ranked among the top 300 students in India eligible for the Indian National Chemistry Olympiad.

2013

National Standard Examinations in Physics (IAPT)

Ranked among the top 1% of students in Rajasthan State in India.

2013

JOURNAL PUBLICATIONS

[J7.] **Arora, P.**, Ceferino, L. and Dobler, G.: A Novel Framework for Detecting Disaster-Induced Power Outages Globally via Satellite Observations, [Preprint], <https://doi.org/10.31224/5148>, 2025

[J6.] **Arora, P.** and Ceferino, L.: The Feasibility of Energy Communities for Hurricane Resilience, [Preprint], <https://doi.org/10.31224/3899>, 2024

- [J5.] **Arora, P.** and Ceferino, L.: A Quasi-Binomial Regression Model for Hurricane-Induced Power Outages during Early Warning, *ASCE-ASME Journal of Risk and Uncertainty Analysis in Engineering Systems*, <https://doi.org/10.1061/AJRUA6.RUENG-1215>, 2024
- [J4.] Lakshmi, K., **Arora, P.** and Rama Mohan Rao, A.: A Global-Local Damage Diagnostic Approach for Large Civil Structures with Very Limited Sensors, *International Journal of Structural Stability and Dynamics*, <https://doi.org/10.1142/S0219455424502705>, 2024
- [J3.] Pachev, B., **Arora, P.**, del-Castillo-Negrete, C. et al.: A Framework for Flexible Peak Storm Surge Prediction, *Coastal Engineering*, <https://doi.org/10.1016/j.coastaleng.2023.104406>, 2023
- [J2.] **Arora, P.** and Ceferino, L.: Probabilistic and Machine Learning Methods for Uncertainty Quantification in Power Outage Prediction due to Extreme Events, *Natural Hazards and Earth System Sciences*, 23, 1665–1683, <https://doi.org/10.5194/nhess-23-1665-2023>, 2023
- [J1.] Singh, M., Yadav, S., Hussain, S., **Arora, P.** and Srivastava, A.: Reliability-Based Mix Design of Marble Dust Incorporated Concrete and Its Assessment Using the Concept of Performance Index, *Structural Concrete*, <https://doi.org/10.1002/suco.202100501>, 2022

CONFERENCE PAPERS AND PRESENTATIONS

- [C4.] **Arora, P.**, Ceferino, L.: A Novel Framework for Detecting Disaster-Induced Power Outages Globally via Satellite Observations and Physics-Guided AI, *ASCE Engineering Mechanics Institute (EMI) Conference*, Anaheim, CA, USA, 2025.
- [C3.] **Arora, P.**, Ceferino, L.: A Performance-Based Probabilistic Framework to Model Risk to Power Systems from Hurricanes, *14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14)*, Dublin, Ireland, 2023.
- [C2.] **Arora, P.**, Ceferino, L.: Could Rooftop Solar Panels and Storage Have Enhanced the Electricity Resilience During Hurricane Isaias (2020)?, *14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14)*, Dublin, Ireland, 2023.
- [C1.] Athira, G., Sanket, R., Aniruddha, T., **Arora, P.**, and Bahurudeen, A.: Assessment of Sustainable Alternative Cementitious Materials Using Characterisation Techniques, *Proceedings of Civil Engineering Conference – Innovation for Sustainability (CEC-2016)*, NIT Hamirpur, India, Sept. 9–10, 2016, pp. 430–445.

TECHNICAL REPORTS

- [T3.] Wolohan, K., Mosalam, K., Jana, A., Bektas, N., Erazo, K., Garcia, S., Kyprioti, A., Wang, C., Romão, X., Petreski, B., Xu, S., Diekmann, A., Dang, H., **Arora, P.**, Lahna, T. (2024). "StEER: 2024 Hualien City Earthquake Annotated Media Repository", in *StEER – 2024 Hualien City, Taiwan Earthquake*. DesignSafe-CI. <https://doi.org/10.17603/ds2-4xv5-qc41>
- [T2.] Cortes, M., **Arora, P.**, Ceferino, L., Ibrahim, H., Istrati, D., Reed, D., Roueche, D., Safiey, A., Tomiczek, T., Zisis, I., Alam, M., Kijewski-Correa, T., Prevatt, D., Robertson, I. (2022). "StEER: Hurricane Ian Preliminary Virtual Reconnaissance Report (PVRR)", in *StEER – Hurricane Ian*. DesignSafe-CI. <https://doi.org/10.17603/ds2-kc9k-s242>
- [T1.] Alam, M., Rodgers, J., Pilkington, S., **Arora, P.**, Gurley, K., Robertson, I. (2022). "StEER: Typhoon Merbok Preliminary Virtual Reconnaissance Report (PVRR)", in *StEER – 17 September 2022 Typhoon Merbok*. DesignSafe-CI. <https://doi.org/10.17603/ds2-4az4-v122>

OPEN-SOURCE DATA AND COMPUTATIONAL RESOURCES

- [O3.] **Arora, P.**, Ceferino, L. (2025). "Power Outage Prediction Utilizing Satellite-based Nightlights Decay," in *Power Outage Prediction Utilizing Satellite-based Nightlights Decay [Version 2]*. DesignSafe-CI. <https://doi.org/10.17603/ds2-a8cv-yc65>
- [O2.] **Arora, P.**, Ceferino, L. (2024). "Hurricane Risk Analysis of Power Failures," in *Hurricane Risk Analysis of Power Systems [Version 4]*. DesignSafe-CI. <https://doi.org/10.17603/ds2-a510-6f83>
- [O1.] Pachev, B., Del-Castillo-Negrete, C., **Arora, P.**, Dawson, C., Valseth, E. (2023). "Reduced Order Models for Peak Storm Surge in Texas and Alaska," in *ADCIRC Reduced Order Modeling*. DesignSafe-CI. <https://doi.org/10.17603/ds2-85j6-k851>

WORKSHOP AND SUMMER SCHOOL PARTICIPATION

University of Washington NSF-Funded RAPID Facility Workshop, July 2025. Participated in the NSF RAPID Workshop focused on advancing field reconnaissance, data collection, and resilience modeling for natural hazard response.

Extremes Workshop, Colorado School of Mines (NSF-Funded), November 2024. Attended the interdisciplinary workshop on modeling and analysis of extreme events; engaged in discussions on uncertainty quantification, resilience assessment, and multi-hazard frameworks.

Uncertainty Quantification Summer School, University of Southern California, August 2024. Completed intensive training on uncertainty modeling, Bayesian inference, and stochastic methods for civil and environmental systems.

DesignSafe Hackathon, Texas Advanced Computing Center (NHERI), July 2022. Developed a neural network-based storm surge forecasting model; collaborated with UT Austin researchers to publish a journal article contributing to the field of natural hazard prediction.

North American Regional Science Council Summer School, July 2022. Selected as one of 12 participants for the program on climate change adaptation, engaged in planning exercises to enhance New York City's resilience to storm and flood events.

TALKS AND OTHER PARTICIPATION

Arora, P. (2024). *Power Outage Modeling and Adoption of Renewables Toward Power Resilience During Extreme Weather Events*. Lightning Talk and Poster presented at the Extremes Workshop, Colorado School of Mines, November 2024.

Ceferino, L., & Arora, P. (2023). *Evaluating the Resilience of Hospitals and Power Systems to Earthquakes and Hurricanes*. Invited Talk presented at Thornton Tomasetti, New York, NY, 2023.

Invited Participant to the culmination of the fellowship project, Sixth International Conference on Disaster Resilient Infrastructure (ICDRI), 2024.

GRANT WRITING EXPERIENCE

NSF FIRE: Fire Science Innovations through Research and Education, submitted, 2025

NSF MATH-DT: Scalable Bayesian Online Learning for Digital Twins in Power Network Resilience against Sequential Extreme Events, awarded, 2025

Bakar Fellows Program, UC Berkeley: OutageIQ-Intelligent Systems for Power Infrastructure Risk Management for Climate and Disaster Resilience, 2025

Pacific Gas & Electric Company: Statistical Modeling of Power Outages in the Humboldt Region, awarded, 2024

New York City Housing Authority: Climate-Induced Power Outage Recurrence Intervals for Resilience Hubs, 2023

Coalition of Disaster Resilient Infrastructure: Resilience of Power Grid Infrastructure with Renewables to Extreme Events in a Changing Climate, Co-PI, awarded, 2022

RESEARCH PROJECTS

Digital Twin for Power System Resilience under Sequential Extreme Events August 2025 - Present
Postdoctoral Research, Advisors - Dr. Luis Ceferino, Dr. Ziqi Wang, and Dr. Michael Lindsey UC Berkeley, CA

- Bayesian updating of structural properties of power poles based on sensor measurements.
- Online updating of the digital twin to optimize the dispatch of repair crews.

Power System Recovery Modeling during Earthquake August 2025 - Present
Postdoctoral Research, Advisor - Dr. Luis Ceferino UC Berkeley, CA

- Agent-based model to identify the recovery sequence of power outages, improving recovery time prediction for customers.
- Modeling the interdependencies of critical infrastructure for comprehensive modeling of power outage recovery.

Hazard Analysis and Urban Utilities Management under Extreme Events January 2021 - August 2025
Ph.D. Project, Advisor - Dr. Luis Ceferino New York University, NY

- Study on the feasibility of adoption of solar panels during hurricanes, involving modeling hurricane hazard, quantification of available solar availability through empirical models, and fragility analysis of power infrastructure.
- Predicting the power outages due to high winds of a hurricane using machine learning methods.
- Utilized remote sensing and AI to develop a power outage model using satellite-based nightlights observation.

Storm Surge Predictions using Neural Networks

Research in Collaboration with UT Austin

August 2022 - Present
New York University, NY

- Understand the importance of hurricane parameters and bathymetry on storm surge predictions.
- Developed a neural network-based storm surge prediction model to significantly reduce computation time compared to high fidelity models such as ADCIRC.
- Develop a global storm surge prediction model to predict storm-related surge in different basins.

Data fusion of LIDAR data and Imagery data for sea waves study

Advisor - Professor Anne Kiremidjian

April 2019 - December 2019
Stanford, CA

- Transforming 2D image data to 3D points based on collinearity condition equations.
- Combining sea wave data from LIDAR, RGB camera, and Infrared camera to identify breaking sea waves.

Structural Health Monitoring

Structural Engineering Research Centre, Advisor - Dr. A Rama Mohan Rao

July 2017 - December 2017
Chennai, India

- Applied method for Damage Identification using Statistical Methods, based on residuals for time-series data using Auto-Regressive Moving Average Model and Kolmogorov Statistical Distance.
- Implemented robust methods to account for variations in SHM data due to different environmental conditions, segregating the outliers in the data from different origins.

TEACHING RESPONSIBILITY

Teaching Assistant

Engineering Mechanics, New York University

August 2023 - December 2023
New York, NY

- Led discussion and tutorial sections for a class of 16 students on structural analysis.
- Conducted weekly office hours to assist graduate students with weekly assignments and class projects.

Teaching Assistant

Disaster Risk Analysis and Urban Systems Resilience, New York University

January 2023 - May 2023
New York, NY

- Developed course material for probabilistic hurricane hazard modeling.
- Led discussion and delivered lectures for a class of 13 students on Disaster Risk Analysis and Urban Systems Resilience.
- Conducted weekly office hours to assist graduate students with weekly assignments and class projects.

Graduate Course Assistant

Introduction to Performance-Based Earthquake Engineering, Stanford University

January 2020 - March 2020
Stanford, CA

- Led discussion sections for a class of 26 students on probabilistic seismic hazard analysis.
- Conducted weekly office hours to assist graduate students with weekly assignments and class projects.

Tutorial Section Leader

Construction Planning and Technology, BITS Pilani

February 2017 - April 2017
Pilani, India

- Delivered a lecture to junior class on designing in Revit Architecture for the Construction Planning and Technology course.

REVIEWER RESPONSIBILITIES

- ASCE-ASME Journal of Risk and Uncertainty Analysis in Engineering Systems
- Journal of Structural Safety and System Reliability
- Frontiers in Energy Research
- ASCE Natural Hazards Review
- International Journal of Disaster Risk Reduction
- Natural Hazards Review

MENTORSHIP

Graduate Mentor for Capstone Project, New York University

January 2023 - May 2023

- Mentored a diverse group of three graduate students for a capstone project at the Center for Urban Science and Progress.

- Team successfully leveraged satellite data and machine learning for power outage predictions during hurricanes.

Summer Undergraduate Research Program, New York University

June 2022 - August 2022

- Mentored two undergraduate students to study a power outage prediction tool and the scalability of solar energy.
- Team leveraged data on real-time power outages and global horizontal solar irradiance.

Mentor to Master's Student, New York University

January 2022 - May 2022

- Mentored graduate students to understand the risks to the power system from hurricanes.
- Team leveraged urban nightlight images to identify power outages.

Seismic Design Competition, Earthquake Engineering Research Institute

October 2021 - May 2022

- Mentored undergraduate students for the seismic design competition at the National Engineering Earthquake Conference.
- Led the team for seismic analysis of a high-rise building using structural software ETABS.

POSITION OF LEADERSHIP

President, Civil Engineering Association, BITS Pilani

January 2018 - May 2018

Publicity Coordinator, Civil Student Society, BITS Pilani

August 2016 - May 2017

CERTIFICATION

FE Civil (NCEES Confirmation ID: 20-166-28)

TECHNICAL SKILLS

Probabilistic Risk Analysis, Catastrophe Modeling, Earthquake and Hurricane Risk Analysis, Machine Learning (Python, R), Statistical Modeling, Geospatial Analysis, ArcGIS, MATLAB, Structural and Infrastructure Risk Evaluation, HPC and distributed computing (OpenMPI, multi-node workflows)

VOLUNTEER ACTIVITIES

- Volunteer, National Service Scheme (NSS), taught high-school students from rural areas near Pilani.
- Student Member, Indian Concrete Institute, August 2016 - August 2017.
- Volunteer, Blood Donation Camp and *Junoon*, a two-day sports meet organized for specially-abled participants.

REFERENCES

1. **Dr. Luis Ceferino**

Assistant Professor, University of California, Berkeley
Email: ceferino@berkeley.edu

2. **Dr. Gregory Dobler**

Associate Professor, University of Delaware
Email: gdobler@udel.edu

3. **Dr. Eirik Valseth**

Associate Professor, Norwegian University of Life Science
Email: eirik.valseth@nmbu.no